

Sharp nodal length on smooth surfaces

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Abstract

We prove that a nonzero real Laplace eigenfunction with eigenvalue $\lambda > 0$ on a fixed smooth closed connected Riemannian surface has nodal length at most $C\sqrt{\lambda}$. Together with the known lower bound, this proves Yau's conjecture in this setting.

1 Introduction and main result

For a Riemannian metric g , let $\Delta_g = \operatorname{div}_g \nabla_g$, and write $Z_u = \{x : u(x) = 0\}$ for the nodal set of a real function u . We prove the following bound for one arbitrary fixed smooth surface.

Theorem 1.1. *Let (M, g) be a closed connected C^∞ Riemannian surface. There is a finite constant $C = C(M, g)$ such that every nonzero real Laplace eigenfunction satisfying*

$$-\Delta_g u = \lambda u, \quad \lambda > 0,$$

obeys

$$\mathcal{H}_g^1(Z_u) \leq C\sqrt{\lambda}.$$

Here $\mathcal{H}_g^1$ is Hausdorff measure for the Riemannian distance. The same constant applies to every eigenfunction in every positive eigenspace.

Yau's conjecture asks for upper and lower bounds of order $\sqrt{\lambda}$ for the $(n - 1)$ -dimensional measure of the nodal set on a smooth closed n -dimensional manifold [13]. Donnelly and Fefferman proved both bounds for real analytic manifolds and metrics [3, Theorem 1.2]. For smooth surfaces, Brüning established the lower bound [1, Satz 2]; Logunov subsequently proved the lower bound in every dimension [7, Theorem 1.1]. For the surface upper bound with a smooth metric, Donnelly and Fefferman proved $C\lambda^{3/4}$ [4], and Dong gave a different proof at the same order [5]. Logunov and Malinnikova then obtained $C\lambda^{3/4-\beta}$ for a universal $\beta \in (0, 1/4)$ [8, Equation (1.1)]. Together with the known lower bound, Theorem 1.1 resolves Yau's conjecture positively for smooth closed surfaces.

Mean growth as the target

Set $k = \sqrt{\lambda}$, the frequency. In local smooth isothermal coordinates the metric is $g = p(x)(dx_1^2 + dx_2^2)$, with $p > 0$ smooth, as in [2, Section 1], and the eigenfunction equation becomes

$$\Delta u + k^2 p(x)u = 0.$$

Nazarov, Polterovich, and Sodin advocated a statistical approach to local doubling and sketched its relation to nodal length, building on Nadirashvili's approach [9, Section 7.3]. Roy-Fortin

established precise comparisons between growth and nodal length [11, 12]. We prove the uniform mean growth bound needed for the sharp upper estimate and use his planar theorem for the final transfer.

Here is why an average suffices. For a coordinate square Q of side s and center x_Q , consider the unweighted logarithmic growth

$$E(Q, 0) = \log \frac{\|u\|_{L^2(\mathbb{D}_{Rs}(x_Q))}}{\|u\|_{L^2(\mathbb{D}_s(x_Q))}},$$

where R is the fixed large disk ratio chosen below and $\mathbb{D}_r(x)$ is the Euclidean disk of radius r . The norms are unnormalized. Subdivide a fixed coordinate square into congruent squares of side s comparable to k^{-1} , stopping once ks is sufficiently small; call these the terminal squares. The estimate proved below gives

$$\text{avg}_{Q \text{ terminal}} E(Q, 0) \leq C.$$

On each such square, Roy-Fortin's theorem, after rescaling and a change from supremum norms to L^2 norms, gives

$$\mathcal{H}^1(Q \cap Z_u) \leq Cs(1 + E(Q, 0)).$$

There are $O(s^{-2})$ squares in a fixed coordinate region. Summing the local estimates therefore gives $O(s^{-1}) = O(k)$ nodal length. The precise grids, radii and constants are specified in Sections 3 and 6. The task is thus to control the average growth while allowing large values on individual squares.

Affine correction and persistent growth

The analytic step extracts logarithmic profiles from norms measured on an exponential scale. With the required separation from the rescaled frequency ks , a localized Carleman estimate makes these profiles subharmonic, even after arbitrarily large affine tilts. Its decisive feature is two-dimensional: when the normalized derivative energy is perpendicular to the weight gradient, the commutator sees the full trace of the weight Hessian.

A local affine part of a profile presents an obstruction to decay: its oscillation and ks decrease by the same factor under subdivision. We subtract that affine part and carry its slope separately. The remaining profile growth decreases faster than the side length on average, while the changes of slope have bounded average cost. Section 3 quantifies both estimates; Section 4 sums their costs along paths that may stop.

To use this improvement through many scales, we divide a grid path into excursions of high growth. A newly entered excursion produces a nearby set of positive area on which high growth persists to the terminal scale. These witness sets may lie outside their entry squares. A packing argument controls them by counting discrepancies between spatial membership and grid ancestry; those discrepancies are confined to summable boundary collars. This geometric argument is stated separately in Lemma 5.4. It permits the analytic persistence statement to be used without requiring its witnesses to lie inside the original squares.

Section 2 constructs the subharmonic limits and establishes the initial growth bound. Section 3 proves the affine subdivision estimate, and Section 4 controls stopped descents and persistent slopes. Section 5 proves persistence at entry, packs the excursions, and bounds the terminal mean growth. Section 6 completes the nodal-length estimate. All coordinate constructions are local, so no orientability assumption is needed.

2 Subharmonic limits of tilted norms

We record exponential growth through logarithms of squared local L^2 norms. Pointwise logarithms are singular at zeros; the compactness statement below instead controls masses on all compact and open sets, which is what later comparisons of norms require. Its compatibility with affine weights will allow us to change the tilt as we subdivide. Throughout the planar arguments, norms and integrals are Euclidean and are not normalized by area. We write $\mathbb{D}_r(a)$ for the open Euclidean disk of radius r centered at a , and $\mathbb{D}_r = \mathbb{D}_r(0)$; a bar denotes closure. For a measurable planar set E , the notation $|E|$ denotes its Euclidean area.

Lemma 2.1 (Extraction of a logarithmic profile). *Let X be an open subset of $\mathbb{R}^2$ or a compact smooth surface. Suppose that $S_j \rightarrow \infty$ and that μ_j are nonnegative Borel measures on X with $\mu_j(X) \leq 1$. There are a subsequence and an upper semicontinuous function $V: X \rightarrow [-\infty, 0]$ such that, for every compact $F \subset X$ and every open $G \subset X$,*

$$\limsup_{j \rightarrow \infty} \frac{\log \mu_j(F)}{2S_j} \leq \sup_F V, \quad \liminf_{j \rightarrow \infty} \frac{\log \mu_j(G)}{2S_j} \geq \sup_G V. \quad (2.1)$$

Here $\log 0 = \sup \emptyset = -\infty$. More generally, if continuous real functions f_j converge locally uniformly to f , then

$$\begin{aligned} \limsup_{j \rightarrow \infty} \frac{1}{2S_j} \log \int_F e^{2S_j f_j} d\mu_j &\leq \sup_F (V + f), \\ \liminf_{j \rightarrow \infty} \frac{1}{2S_j} \log \int_G e^{2S_j f_j} d\mu_j &\geq \sup_G (V + f). \end{aligned}$$

The same subsequence works for all these sets and weights.

Proof. Choose a countable basis $\{G_i\}_{i \geq 1}$ consisting of relatively compact open sets. A diagonal subsequence makes all the limits

$$\ell_i = \lim_{j \rightarrow \infty} \frac{\log \mu_j(G_i)}{2S_j} \in [-\infty, 0]$$

exist. Define $V(x) = \inf_{i: x \in G_i} \ell_i$. If $V(x) < a$, one of these basis neighborhoods has $\ell_i < a$, and $V < a$ throughout that neighborhood. Thus V is upper semicontinuous.

For any finite $a > \sup_F V$, every point of F has a basis neighborhood with exponent less than a . A finite subcover and $\mu_j(F) \leq \sum_i \mu_j(G_i)$ give the compact upper bound by a . Letting $a \downarrow \sup_F V$ proves the first inequality, also when this supremum is $-\infty$. If $x \in G$, a basis neighborhood G_i with $x \in G_i \subset G$ gives a lower bound by $\ell_i \geq V(x)$, proving the second inequality.

For the weighted upper bound, take $a > \sup_F (V + f)$ and choose the neighborhood at each point small enough that its limiting measure exponent plus the supremum of f on it is less than a . This is possible by continuity of f and the definition of V , including at points where $V = -\infty$. A finite cover and uniform convergence of f_j on its compact closure give the claimed upper bound. For the lower bound at a point $x \in G$ with $V(x) > -\infty$, choose G_i compactly contained in G so small that $f \geq f(x) - \varepsilon$ there. The weighted integral over G_i has lower exponent at least $\ell_i + f(x) - \varepsilon$. Let $\varepsilon \downarrow 0$ and then take the supremum over x . $\square$

We call V a *logarithmic profile*. Multiplying the measures by positive densities uniformly bounded above and below on compact sets does not change the profile bounds: the logarithms of these bounds disappear after division by $2S_j$. We may also use the compact upper bound for a relatively compact set by taking its closure. Normalizing mass on an open disk alone does not assert that the profile has a finite value inside the disk; the identically $-\infty$ case remains possible.

Theorem 2.2 (Subharmonicity of logarithmic profiles). *On a fixed planar disk Ω , let real smooth functions U_j, p_j satisfy*

$$\Delta U_j + K_j^2 p_j U_j = 0,$$

where $K_j \geq 0$. Let $S_j > 0$, $B_j \in \mathbb{R}^2$, and $D_j = S_j + |B_j|$. Assume

$$\begin{aligned} S_j &\longrightarrow \infty, & \frac{K_j}{D_j} &\longrightarrow 0, \\ \sup_j \|p_j\|_\infty &< \infty, & \frac{K_j^2 \|\nabla p_j\|_\infty}{S_j D_j} &\longrightarrow 0. \end{aligned} \tag{2.2}$$

Local versions of the coefficient assumptions on relatively compact subdisks suffice. For positive constants c_j , suppose the measures

$$d\mu_j = c_j e^{-2B_j \cdot x} |U_j(x)|^2 dx$$

have mass at most one on Ω . Every upper semicontinuous logarithmic profile with the compact, open, and weighted bounds of Lemma 2.1 is subharmonic on Ω , with the identically $-\infty$ function allowed.

The main estimate concerns weights whose normalized gradient is nonzero. It is essential that the next lemma allows D_j/S_j to be arbitrarily large.

Lemma 2.3 (A local Carleman estimate). *Assume the coefficient and parameter hypotheses of Theorem 2.2. After passing to a subsequence, write*

$$S_j/D_j \rightarrow d, \quad B_j/D_j \rightarrow b, \quad d \geq 0, \quad d + |b| = 1.$$

Let χ be a real smooth function near $x_0 \in \Omega$ such that $\Delta \chi(x_0) < 0$ and $b + d \nabla \chi(x_0) \neq 0$. Put $T_j(x) = -B_j \cdot x - S_j \chi(x)$. There are a neighborhood Ω_0 of x_0 , a constant $c > 0$, and an index j_0 such that

$$\|e^{T_j}(\Delta + K_j^2 p_j)w\|_2 \geq c \sqrt{S_j} D_j \|e^{T_j} w\|_2 \tag{2.3}$$

for all $j \geq j_0$ and all real $w \in C_c^\infty(\Omega_0)$.

The two parameters reflect different derivatives of the weight: $\nabla T_j = -B_j - S_j \nabla \chi$ has scale D_j , whereas $T_j'' = -S_j \chi''$ has scale S_j . The affine tilt can therefore enlarge the gradient without enlarging the curvature.

Proof. Suppress the index and write $v = e^T w$. The conjugated operator is $P_+ + P_-$, where

$$P_+ = \Delta + |\nabla T|^2 + K^2 p, \quad P_- = -2\nabla T \cdot \nabla - \Delta T.$$

On compactly supported functions, P_+ is symmetric and P_- is skew-symmetric. Integration by parts gives

$$\begin{aligned} 2\langle P_+ v, P_- v \rangle &= 4 \int T''(\nabla v, \nabla v) \\ &\quad + \int (4T''(\nabla T, \nabla T) - \Delta^2 T + 2K^2 \nabla p \cdot \nabla T) v^2. \end{aligned} \tag{2.4}$$

For clarity, the contribution from Δ is $4 \int T''(\nabla v, \nabla v) - \int \Delta^2 T v^2$. Multiplication by a real function a contributes $2 \int \nabla a \cdot \nabla T v^2$; applying this with $a = |\nabla T|^2 + K^2 p$ gives the remaining terms in Equation (2.4).

If Equation (2.3) failed on every sufficiently small neighborhood, a diagonal choice would give a subsequence and real smooth v_j with supports shrinking to x_0 , such that

$$\|v_j\|_2 = 1, \quad \|(P_+ + P_-)v_j\|_2 = o(\sqrt{S_j D_j}).$$

All constants below are uniform on one fixed small neighborhood of x_0 . There, $|\nabla T| \leq CD$, $|T''| + |\Delta^2 T| \leq CS$, and the last hypothesis in Equation (2.2) controls the potential term in Equation (2.4). Set $X = \|P_+ v\|_2$, $Y = \|P_- v\|_2$, and $G = \|\nabla v\|_2$. Expanding the squared norm and using the identity gives

$$X^2 + Y^2 \leq CS(G^2 + D^2).$$

Testing P_+ against v gives $G^2 \leq CD^2 + X$, since $K = o(D)$. Combining the two inequalities first yields $X \leq C(S + \sqrt{SD})$ and then

$$G \leq CD, \quad X + Y \leq C\sqrt{SD}.$$

Here $D \geq S \rightarrow \infty$ absorbs the term S .

Uniformly on the shrinking supports,

$$\nabla T_j / D_j \rightarrow a = -b - d\nabla \chi(x_0) \neq 0, \quad T_j'' / S_j \rightarrow H = -\chi''(x_0).$$

The preceding bounds give

$$\frac{\sqrt{S_j}}{D_j} \leq \frac{1}{\sqrt{S_j}} \rightarrow 0.$$

Dividing the identity obtained by testing P_+ against v_j by D_j^2 therefore shows

$$\int |\nabla v_j / D_j|^2 \rightarrow |a|^2.$$

Likewise,

$$\left\| \frac{\nabla T_j}{D_j} \cdot \frac{\nabla v_j}{D_j} \right\|_2 \leq \frac{\|P_- v_j\|_2 + \|\Delta T_j v_j\|_2}{2D_j^2} \rightarrow 0.$$

Replacing $\nabla T_j / D_j$ by its uniform limit consequently gives $\|a \cdot \nabla v_j / D_j\|_2 \rightarrow 0$.

Let e be a unit vector perpendicular to a . In dimension two, the energy perpendicular to a has only this one direction. Decomposing $\nabla v_j / D_j$ in the orthonormal basis $\{a/|a|, e\}$, the parallel component tends to zero in L^2 , whereas the squared norm of the perpendicular component tends to $|a|^2$. Hence the first term in Equation (2.4), divided by $S_j D_j^2$, tends to $4|a|^2 H(e, e)$. The other main Hessian term tends to $4H(a, a)$. The two remaining terms tend to zero: their absolute values after this division are bounded by

$$\frac{C}{D_j^2} \quad \text{and} \quad \frac{CK_j^2 \|\nabla p_j\|_\infty}{S_j D_j},$$

respectively. We have thus proved

$$\begin{aligned} \frac{2\langle P_+ v_j, P_- v_j \rangle}{S_j D_j^2} &\rightarrow 4(|a|^2 H(e, e) + H(a, a)) \\ &= 4|a|^2 \operatorname{tr} H > 0. \end{aligned}$$

The squared norms of P_+v_j and P_-v_j are nonnegative, so this contradicts the assumed smallness of $\|(P_+ + P_-)v_j\|_2^2$. $\square$

We also need to remove one possible exceptional gradient from the upper-test criterion. The following elementary perturbation does so without any initial integrability assumption. An upper test at a finite point x_0 is a smooth function agreeing with the tested function at x_0 and dominating it in a neighborhood of x_0 . This is an upper-semicontinuous subharmonic variant of the standard viscosity exceptional-gradient-removal principle; compare [6, Section 5, Lemma 6]. We give the local proof for functions allowed to take the value $-\infty$, rather than apply that lemma directly.

Lemma 2.4 (Removal of the exceptional gradient). *Let $W : \Omega \rightarrow [-\infty, \infty)$ be upper semicontinuous and locally bounded above on a connected planar domain. Suppose every smooth function ψ touching W from above at a finite point x_0 , with $\nabla\psi(x_0) \neq 0$, satisfies $\Delta\psi(x_0) \geq 0$. Then W is subharmonic, with the identically $-\infty$ function allowed.*

Proof. Suppose first that an upper test at 0 has zero gradient and negative Laplacian; translate the point to 0 if necessary. A quadratic Taylor polynomial with its Hessian increased by a sufficiently small positive multiple of the identity gives a quadratic upper test h with

$$\begin{aligned} W(0) &= h(0), & \nabla h(0) &= 0, & \operatorname{tr} h'' &< 0, \\ W(x) &< h(x) & (0 < |x| &\leq r) \end{aligned}$$

for some small $r > 0$. Choose a unit eigenvector e of h'' with eigenvalue $\alpha < 0$. For small $t > 0$, maximize $W(x) - h(x) + te \cdot x$ on $\overline{\mathbb{D}_r}$. Upper semicontinuity and the strict boundary gap ensure that a maximizer x_t is interior. Its value is at least the value 0 at the origin, and $W - h \leq 0$, so $e \cdot x_t \geq 0$. At x_t an upper test is $h(x) - te \cdot x$, shifted by its contact constant. Its gradient has component

$$e \cdot \nabla(h - te \cdot x)(x_t) = \alpha e \cdot x_t - t < 0,$$

while its Laplacian is still negative. This contradicts the hypothesis. All smooth upper tests at finite points therefore have nonnegative Laplacian.

We obtain harmonic comparison directly. Let $\overline{\mathbb{D}_r(a)} \subset \Omega$, let h be harmonic in $\mathbb{D}_r(a)$ and continuous on its closure, and suppose $W \leq h$ on the boundary. If $W - h$ were positive somewhere inside, then for sufficiently small $\varepsilon > 0$ the function

$$h_\varepsilon(x) = h(x) + \varepsilon(r^2 - |x - a|^2)$$

would still lie below W at some interior point. A positive maximum of $W - h_\varepsilon$ is attained at an interior finite point. Adding this maximum to h_ε supplies an upper test with Laplacian $-4\varepsilon < 0$, again a contradiction. Thus W satisfies harmonic comparison. For upper semicontinuous functions this is the usual characterization of subharmonicity, including the identically $-\infty$ case; see [10, Chapter 2]. In particular, local integrability and the submean property of a nontrivial subharmonic function are consequences here, not assumptions used in the argument. $\square$

Proof of Theorem 2.2. Fix such a profile V and pass to a further subsequence so that $S_j/D_j \rightarrow d$ and $B_j/D_j \rightarrow b$ as in Lemma 2.3. The profile bounds remain valid on this subsequence. Suppose a smooth χ touches V from above at a finite point x_0 , and suppose

$$\Delta\chi(x_0) < 0, \quad b + d\nabla\chi(x_0) \neq 0.$$

By adding a small positive multiple of $|x - x_0|^4$, we make the touching strict on a sufficiently small closed disk without altering the gradient or Hessian at x_0 . Take this disk inside the neighborhood in Lemma 2.3.

Choose fixed radii $0 < r_0 < r_1 < r_2 < r_3$ inside that neighborhood. Let ζ be a smooth cutoff equal to one on $\mathbb{D}_{r_1}(x_0)$ and supported in $\mathbb{D}_{r_2}(x_0)$. Its derivatives are supported in the annulus between these two disks. Set

$$G = \mathbb{D}_{r_0/2}(x_0), \quad F = \{x : r_0 \leq |x - x_0| \leq r_3\}.$$

We may choose the radii so that the strict touching holds on the whole closed outer disk. Then F is a compact annulus, $\sup_F(V - \chi) < 0$, and $\sup_G(V - \chi) = 0$.

For completeness, the cutoff error can be controlled with a prefactor independent of the ratio D_j/S_j . Choose a smooth θ supported in the interior of F and equal to one near the derivatives of ζ . Testing the equation for U_j against $\theta^2 e^{2T_j} U_j$ and integrating by parts bounds

$$\begin{aligned} \int \theta^2 e^{2T_j} |\nabla U_j|^2 &\leq C \int_F e^{2T_j} (|\nabla \theta|^2 + \theta^2 |\nabla T_j|^2 + K_j^2 |p_j|^2 \theta^2) |U_j|^2 \\ &\leq C D_j^2 \int_F e^{2T_j} |U_j|^2. \end{aligned}$$

The first inequality absorbs half of the gradient integral by Cauchy–Schwarz; the second uses $K_j = o(D_j)$ and $D_j \rightarrow \infty$. Since

$$(\Delta + K_j^2 p_j)(\zeta U_j) = (\Delta \zeta) U_j + 2 \nabla \zeta \cdot \nabla U_j,$$

Lemma 2.3 gives

$$c \sqrt{S_j} D_j \|e^{T_j} U_j\|_{L^2(G)} \leq C D_j \|e^{T_j} U_j\|_{L^2(F)}.$$

Cancel D_j before comparing exponential rates. Squaring and multiplying by the common normalization constant c_j yields

$$S_j \int_G e^{-2S_j \chi} d\mu_j \leq C \int_F e^{-2S_j \chi} d\mu_j.$$

The weighted bounds of Lemma 2.1, together with $\log S_j/S_j \rightarrow 0$, now give the contradiction

$$0 = \sup_G (V - \chi) \leq \sup_F (V - \chi) < 0.$$

In particular, no bound on $\log D_j/S_j$ was required.

We have established the upper-test inequality whenever $b + d \nabla \chi(x_0) \neq 0$. If $d = 0$, then $|b| = 1$ and there is no exceptional gradient; harmonic comparison follows as in Lemma 2.4. If $d > 0$, put $W(x) = V(x) + (b/d) \cdot x$. An upper test ψ for W corresponds to the upper test $\chi = \psi - (b/d) \cdot x$ for V , and $b + d \nabla \chi = d \nabla \psi$. Thus W satisfies the hypothesis of Lemma 2.4. It is subharmonic, and subtracting the affine function shows that V is subharmonic too. $\square$

The next estimate is a fixed-disk form of quantitative unique continuation. For the classical doubling estimate of Donnelly and Fefferman and the corresponding propagation bound, see [9, Theorem 4.1 and Lemma 4.14]. We give a proof using logarithmic profiles, which also supplies the local nonvanishing needed below.

Proposition 2.5 (Initial growth on a fixed disk). *Let (M, g) be a fixed smooth closed connected surface, and let $G \subset M$ be a fixed nonempty open coordinate disk. There is a finite constant $C = C(M, g, G)$ such that every nonzero real eigenfunction $-\Delta_g u = k^2 u$ with $k > 0$ satisfies $\|u\|_{L^2(G)} > 0$ and*

$$\log \frac{\|u\|_{L^2(M)}}{\|u\|_{L^2(G)}} \leq Ck. \quad (2.5)$$

Both norms use Riemannian volume. The constant is uniform over all eigenfunctions in every eigenspace.

Proof. The positive spectral gap gives $k \geq k_1 > 0$. Interpret the log ratio in Equation (2.5) as $+\infty$ when the local norm is zero. If the assertion failed, we could choose eigenfunctions u_j , normalized by $\|u_j\|_{L^2(M)} = 1$, such that their log ratios R_j obey $R_j \geq j^2 k_j$. This also covers a zero local norm by repeating that eigenfunction. Set $S_j = j k_j$. Then $S_j \rightarrow \infty$ and $S_j/k_j \rightarrow \infty$, whether or not k_j tends to infinity.

Apply Lemma 2.1 on M to the probability measures $|u_j|^2 d\text{vol}_g$. Compactness of M and the compact upper bound with $F = M$ give $0 \leq \sup_M V \leq 0$. Upper semicontinuity ensures that this maximum is attained.

In a relatively compact isothermal chart, $g = p(x)(dx_1^2 + dx_2^2)$ with $p > 0$ smooth, and the coordinate functions satisfy $\Delta U_j + k_j^2 p U_j = 0$. The positive volume factor p is bounded above and below on this chart. Multiplying $|U_j|^2 dx$ by a fixed positive constant if necessary makes its mass at most one; the resulting logarithmic profile is the restriction of V . Theorem 2.2 applies with $B_j = 0$, $K_j = k_j$, and $D_j = S_j$, since

$$\frac{k_j}{S_j} = \frac{1}{j} \rightarrow 0, \quad \frac{k_j^2 \|\nabla p\|_\infty}{S_j^2} = \frac{\|\nabla p\|_\infty}{j^2} \rightarrow 0.$$

Thus V is subharmonic in every such chart. The strong maximum principle makes the set where $V = 0$ open; it is also closed by upper semicontinuity and nonempty by the preceding maximum argument. Connectedness gives $V \equiv 0$ on M .

On the other hand, the normalized mass of the given disk satisfies

$$\frac{1}{2S_j} \log \int_G |u_j|^2 d\text{vol}_g = -\frac{R_j}{S_j} \leq -j,$$

with the same conclusion when the integral is zero. This contradicts the open lower bound in Equation (2.1), whose right-hand side is $\sup_G V = 0$. The argument proves both positivity of the local norm and Equation (2.5). It uses only local isothermal charts and therefore imposes no orientability assumption. $\square$

3 Affine improvement under subdivision

The subharmonic profiles of Section 2 admit a useful local affine correction. Averaged over a sufficiently fine square grid, the remaining growth decreases faster than the side length. We first establish this fact for profiles and then transfer it to eigenfunctions.

Definition 3.1 (Growth and affine excess). Fix an isothermal coordinate region in which $\Delta u + k^2 p u = 0$, and work in a relatively compact subset on which p and ∇p have fixed uniform bounds. Set $R = 1000$. Choose a root square Q_0 , with side $s_0 \leq 1$ and center x_{Q_0} , such that $\overline{\mathbb{D}_{2Rs_0}(x_{Q_0})}$ lies in this subset. For an integer $A \geq 2$, subdivide each square into A^2 congruent squares and write $q \prec Q$ when q is an immediate child of Q . For a square Q of side $s = s(Q)$ and center x_Q , put $K_Q = ks$. For a real vector $b \in \mathbb{R}^2$, define

$$\begin{aligned} E(Q, b) &= \log \frac{\|e^{-kb \cdot (x - x_Q)} u\|_{L^2(\mathbb{D}_{Rs}(x_Q))}}{\|e^{-kb \cdot (x - x_Q)} u\|_{L^2(\mathbb{D}_s(x_Q))}}, \\ m(Q, b) &= \frac{E(Q, b)}{K_Q}, \quad N(Q) = m(Q, 0). \end{aligned} \tag{3.1}$$

All planar norms here and below are unnormalized Euclidean norms. An average over children assigns weight A^{-2} to each child.

A square Q is called *admissible* if $s(Q) \leq 1$ and $\overline{\mathbb{D}_{Rs(Q)}(x_Q)}$ lies in the fixed coordinate region where the coefficient bounds hold. All descendants of the chosen root are admissible. The stronger $2R$ margin at the root reserves room for nearby squares outside its subdivision tree.

Proposition 2.5 ensures that the local norms in (3.1) are nonzero. In particular, E , m , and N are finite and nonnegative, since the disks are nested. Comparing the exponential weight on the two disks gives

$$N(Q) \leq m(Q, b) + (R + 1)|b|. \tag{3.2}$$

Indeed the outer untilted norm is at most $e^{KR|b|}$ times its tilted norm, whereas the inner untilted norm is at least $e^{-K|b|}$ times its tilted norm, where $K = K_Q$. The same definitions and estimates apply to finite subtrees of an extended translated grid consisting of admissible squares.

The normalization by K_Q explains why subdivision alone is insufficient. Put $\rho = A^{-1}$. An affine profile with slope $c \neq 0$ has a difference of suprema equal to $(R - 1)\rho|c|$ between concentric disks of radii $R\rho$ and ρ . Division by the child frequency $K_q = \rho K_Q$ cancels this first-order gain. We will subtract the local affine part to obtain an average profile excess of order $\rho^2(1 + |\log \rho|)$, while keeping the average correction slope bounded. The latter bound controls the cost of changing weights through successive subdivisions.

Lemma 3.2 (Uniform potential decomposition). *Let V be subharmonic on $\mathbb{D}_R$, with*

$$V \leq 0, \quad \sup_{\overline{\mathbb{D}_1}} V \geq -1.$$

Then $\|V\|_{L^1(\mathbb{D}_8)}$ and the mass of its Riesz measure $(2\pi)^{-1}\Delta V$ on $\mathbb{D}_5$ are uniformly bounded. There are a nonnegative measure ν on $\mathbb{D}_4$, of uniformly bounded mass, and a harmonic function h on $\mathbb{D}_3$ such that

$$V(x) = h(x) + \int_{\mathbb{D}_4} \log|x - z| d\nu(z), \quad x \in \mathbb{D}_3.$$

The first and second derivatives of h on $\mathbb{D}_2$ are uniformly bounded. All bounds are independent of V , and the representation holds for the pointwise subharmonic representatives, including values equal to $-\infty$.

Proof. Upper semicontinuity supplies a point $x_* \in \overline{\mathbb{D}_1}$ with $V(x_*) \geq -1$. The disk $\mathbb{D}_{10}(x_*)$ contains $\mathbb{D}_8$ and is contained in $\mathbb{D}_R$. In particular, V is nontrivial and locally integrable. Since

$V \leq 0$, the submean inequality gives

$$\int_{\mathbb{D}_8} |V| \leq - \int_{\mathbb{D}_{10}(x_*)} V \leq 100\pi.$$

Its distributional Laplacian is a nonnegative Radon measure. Choose a fixed nonnegative smooth cutoff ψ , supported in $\mathbb{D}_8$ and equal to one on $\mathbb{D}_5$. If $\eta = (2\pi)^{-1}\Delta V$, then

$$\eta(\mathbb{D}_5) \leq \int \psi d\eta = \frac{1}{2\pi} \int V \Delta \psi \leq C \|V\|_{L^1(\mathbb{D}_8)}.$$

Let ν be the restriction of η to $\mathbb{D}_4$, and set $P(x) = \int_{\mathbb{D}_4} \log|x - z| d\nu(z)$. The elementary bound

$$\sup_{z \in \mathbb{D}_4} \int_{\mathbb{D}_3} |\log|x - z|| dx < \infty$$

shows that $\|P\|_{L^1(\mathbb{D}_3)} \leq C\nu(\mathbb{D}_4)$. The logarithmic fundamental solution identity $\Delta \log|x - z| = 2\pi\delta_z$ implies $\Delta(V - P) = 0$ on $\mathbb{D}_3$. Thus $V - P$ agrees almost everywhere with a harmonic function h . Interior harmonic estimates, applied to its bounded $L^1(\mathbb{D}_3)$ norm, bound ∇h and $\nabla^2 h$ on $\mathbb{D}_2$. Finally, the logarithmic potential of a nonnegative measure has its usual subharmonic representative. Both V and $h + P$ are recovered pointwise as limits of their disk averages. Their almost-everywhere equality therefore gives the asserted pointwise representation. $\square$

Lemma 3.3 (Affine correction of a profile). *Let V , h , and ν be as in Lemma 3.2. Subdivide the unit square $[-1/2, 1/2]^2$ into A^2 squares, where $A > 10R$, put $\rho = A^{-1}$, and denote their centers by y . Define*

$$c_y = \nabla h(y) + \int_{\substack{z \in \mathbb{D}_4 \\ |z - y| > 2R\rho}} \frac{y - z}{|y - z|^2} d\nu(z).$$

Then

$$\begin{aligned} & \sup_{\mathbb{D}_{R\rho}(y)} (V(x) - c_y \cdot x) - \sup_{\mathbb{D}_\rho(y)} (V(x) - c_y \cdot x) \\ & \leq C \left(\rho^2 + \int_{\mathbb{D}_4} \min \left\{ 1, \frac{\rho^2}{|z - y|^2} \right\} d\nu(z) \right). \end{aligned} \tag{3.3}$$

At $z = y$ the minimum in this formula is interpreted as one. Moreover,

$$\text{avg}_y |c_y| \leq C, \quad \text{avg}_y [\text{right-hand side of (3.3)}] \leq C\rho^2(1 + |\log \rho|).$$

The constants may depend on R , but are independent of A and V .

Proof. All disks $\overline{\mathbb{D}_{R\rho}(y)}$ lie in $\mathbb{D}_2$. The vector c_y is finite because its kernel is truncated away from its singularity. Put $W_y(x) = V(x) - c_y \cdot x$, and lower bound its supremum on the inner disk by its disk average there. For the harmonic term, this average is its value at y , after the affine part is accounted for. Taylor's theorem bounds the corresponding outer deviation by $C\rho^2$.

For a far source $|z - y| > 2R\rho$, the logarithm is harmonic on the outer disk and its Hessian is bounded there by $C|z - y|^{-2}$. Its inner disk average is $\log|y - z|$, and the retained gradient in c_y removes its first-order term. Its contribution to the outer deviation is therefore at most $C\rho^2/|z - y|^2$.

For a near source $|z - y| \leq 2R\rho$, rescale by ρ . Writing $t = (z - y)/\rho$, the disk-average formula

$$\frac{1}{\pi} \int_{\mathbb{D}_1} \log|\xi - t| d\xi = \begin{cases} (|t|^2 - 1)/2, & |t| \leq 1, \\ \log|t|, & |t| \geq 1 \end{cases}$$

has lower bound $-1/2$. On the larger disk the corresponding logarithm is at most $\log(3R)$, since $|t| \leq 2R$. The common term $\log \rho$ cancels between these two quantities. Thus the difference between the outer upper bound and the inner average is at most $\log(3R) + 1/2$, uniformly in the source position. This calculation permits an atom at y , or anywhere else in the inner disk: the logarithm remains integrable in the averaging variable. At an outer point where a potential is $-\infty$, an upper bound causes no difficulty. Because the measure is nonnegative, the kernel bounds may be integrated. On the near region,

$$\min \left\{ 1, \frac{\rho^2}{|z - y|^2} \right\} \geq \frac{1}{4R^2}.$$

Combining the near and far bounds proves (3.3).

For completeness, fix any $z \in \mathbb{D}_4$. The number of grid centers within distance r of z is at most $C(1 + r^2/\rho^2)$. There are ρ^{-2} centers in total. The centers at distance at most ρ contribute $C\rho^2$ to the averaged minimum kernel. In each dyadic annulus $2^\ell \rho < |z - y| \leq 2^{\ell+1} \rho$, the number of centers is at most $C2^{2\ell}$, whereas the kernel is at most $2^{-2\ell}$. All distances are bounded by five, so only $C(1 + |\log \rho|)$ annuli occur. Hence, uniformly in z ,

$$\text{avg}_y \min \left\{ 1, \frac{\rho^2}{|z - y|^2} \right\} \leq C\rho^2(1 + |\log \rho|).$$

The same annuli, for the truncated gradient kernel, give

$$\rho^2 \sum_{\substack{y \\ |z-y| > 2R\rho}} \frac{1}{|z - y|} \leq C.$$

Indeed an annulus of radius comparable to $2^\ell \rho$ contributes at most $C2^\ell \rho$, and these radii sum to a fixed constant. Integrating the two bounds against the bounded measure ν , and using the bound for ∇h , proves both averaged estimates. $\square$

Proposition 3.4 (One subdivision step). *There are an integer $A > 10R$ and constants $C_0, C_1 > 0$, depending only on the fixed coordinate coefficient bounds and R , with the following property. Let Q be an admissible square of side $s \leq 1$, let $K = ks > 0$, and let $b \in \mathbb{R}^2$ and $S > 0$. Put $B = Kb$. If*

$$S \geq C_0, \quad E(Q, b) \leq S, \quad \frac{K}{S + |B|} \leq \frac{1}{C_0}, \quad \frac{K^2 s}{S(S + |B|)} \leq \frac{1}{C_0}, \quad (3.4)$$

then real vectors b_q can be chosen for the children $q \prec Q$ so that

$$\text{avg}_{q \prec Q} |b_q - b| \leq C_1 \frac{S}{K}, \quad \text{avg}_{q \prec Q} m(q, b_q) \leq \frac{1}{2} \frac{S}{K}. \quad (3.5)$$

The constants are independent of the eigenfunction, the scale, the depth of a tree, and any lower bound for K .

Proof. First choose $A > 10R$, using only the uniform estimates of Lemma 3.3, so large that, with $\rho = A^{-1}$,

$$C\rho^2(1 + |\log \rho|) < \frac{\rho}{4}.$$

Choose C_1 strictly larger than the uniform bound for $\text{avg}_y |c_y|$ in that lemma. We prove that some finite C_0 works for these fixed choices.

Suppose otherwise. There is a sequence of failures of the conclusion for which the hypotheses hold with $C_0 = j$. Use subscripts j for the associated quantities and rescale each parent square to the unit square centered at zero:

$$U_j(X) = u_j(x_{Q_j} + s_j X), \quad p_j(X) = p(x_{Q_j} + s_j X).$$

The rescaled equation is $\Delta U_j + K_j^2 p_j U_j = 0$, and $\|p_j\|_\infty \leq C$, $\|\nabla p_j\|_\infty \leq C s_j$. With $B_j = K_j b_j$ and $D_j = S_j + |B_j|$, the failure hypotheses imply

$$S_j \longrightarrow \infty, \quad \frac{K_j}{D_j} \longrightarrow 0, \quad \frac{K_j^2 \|\nabla p_j\|_\infty}{S_j D_j} \longrightarrow 0.$$

No restriction on $|B_j|/S_j$ has been imposed.

Normalize the tilted measures on $\mathbb{D}_R$ by setting

$$d\mu_j(X) = \frac{e^{-2B_j \cdot X} |U_j(X)|^2 dX}{\int_{\mathbb{D}_R} e^{-2B_j \cdot X} |U_j(X)|^2 dX}.$$

Lemma 2.1 and Theorem 2.2 give, after passage to a subsequence, a subharmonic profile V , possibly identically $-\infty$, on $\mathbb{D}_R$. It satisfies $V \leq 0$, while

$$\mu_j(\mathbb{D}_1) = e^{-2E(Q_j, b_j)} \geq e^{-2S_j}.$$

The compact upper bound in (2.1), applied to $\overline{\mathbb{D}_1}$, therefore gives $\sup_{\overline{\mathbb{D}_1}} V \geq -1$. In particular, V is nontrivial and the two preceding lemmas apply.

For each of the fixed, finitely many child centers y , take the vector c_y furnished by Lemma 3.3 for this limiting profile. On the corresponding child q_j , set

$$b_{q_j} = b_j + \frac{S_j}{K_j} c_y.$$

In parent coordinates the additional weight is $e^{-2S_j c_y \cdot X}$. To see the cancellation of recentering constants, write

$$-K_j b_{q_j} \cdot (X - y) = -B_j \cdot X - S_j c_y \cdot X + (B_j + S_j c_y) \cdot y.$$

The last term, as well as the measure normalization and the coordinate Jacobian, cancels in the ratio defining $E(q_j, b_{q_j})$. Consequently this excess is one half the logarithm of

$$\frac{\int_{\mathbb{D}_{R\rho}(y)} e^{-2S_j c_y \cdot X} d\mu_j(X)}{\int_{\mathbb{D}_\rho(y)} e^{-2S_j c_y \cdot X} d\mu_j(X)}.$$

Apply the weighted compact upper bound to the numerator, using $\overline{\mathbb{D}_{R\rho}(y)}$, and the weighted open lower bound to the denominator. This yields

$$\limsup_j \frac{E(q_j, b_{q_j})}{S_j} \leq \sup_{\overline{\mathbb{D}_{R\rho}(y)}} (V(X) - c_y \cdot X) - \sup_{\mathbb{D}_\rho(y)} (V(X) - c_y \cdot X).$$

The inner supremum is finite: a nontrivial subharmonic function is locally integrable, so it cannot equal $-\infty$ throughout an open disk, and it is bounded above on compact subsets. Both suprema in the displayed difference are therefore finite.

There are only A^2 children, and A has already been fixed. Averaging the last inequality and invoking Lemma 3.3 gives

$$\limsup_j \operatorname{avg}_{q_j \prec Q_j} \frac{E(q_j, b_{q_j})}{S_j} < \frac{\rho}{4}.$$

Since $K_{q_j} = \rho K_j$, it follows, for all sufficiently large j , that

$$\operatorname{avg}_{q_j \prec Q_j} m(q_j, b_{q_j}) \leq \frac{1}{2} \frac{S_j}{K_j}.$$

At the same time,

$$\operatorname{avg}_{q_j \prec Q_j} |b_{q_j} - b_j| = \frac{S_j}{K_j} \operatorname{avg}_y |c_y| \leq C_1 \frac{S_j}{K_j}.$$

Thus these choices satisfy both required inequalities, contrary to the defining failure of the sequence. This proves the proposition. In particular, A and C_1 have been fixed before C_0 , and no positive lower bound for K was used. $\square$

4 Descent and stability

We iterate the subdivision estimate while the untilted growth or the tilt is large. The first goal is a bound, independent of the number of levels, for the total residual growth and the accumulated changes of tilt. This bound will show that a sufficiently large initial tilt with small residual growth forces high untilted growth to persist on a positive fraction of paths.

Fix the integer A and constants C_0, C_1 from Proposition 3.4. We may increase C_0 so that $C_0 \geq 1$. Consider a finite subdivision tree with all terminal squares at the same level, and suppose that

$$K_Q = ks(Q) \geq a_0 > 0$$

on every square of the tree. The square at the start of a subtree will be denoted by I , and its side satisfies $s(I) \leq 1$. All disks used below are assumed to lie in the coordinate region of Section 3. The arguments also apply to a subtree in an extension of the same grid whenever its disks have this property. Throughout this section, constants may depend on the fixed number a_0 , as well as on the coordinate data and the constants already chosen.

Choosing a child uniformly among the A^2 children at each step defines a random path through the tree. Equivalently, one may choose a point uniformly in I and follow its containing squares, ignoring grid boundaries of area zero.

Lemma 4.1 (One step of descent). *There is a fixed $L > 0$ such that, at any nonterminal square Q , if $N(Q) > L$ or $|b| > L$, one can assign a tilt b_q to each child q so that, writing $m = m(Q, b)$, $m' = m(q, b_q)$, and $b' = b_q$, one has*

$$\begin{aligned} \mathbb{E}(|b' - b| \mid Q, b) &\leq C_1(m + f_Q), & \mathbb{E}(m' \mid Q, b) &\leq \frac{1}{2}(m + f_Q), \\ f_Q &= \frac{C_0}{K_Q} + s(Q). \end{aligned} \tag{4.1}$$

Moreover, if $Q_0 = I, Q_1, \dots, Q_d$ is any path from I to the terminal level, then

$$\sum_{i=0}^d f_{Q_i} \leq \frac{A}{A-1} \left(\frac{C_0}{a_0} + s(I) \right). \quad (4.2)$$

Proof. Write $K = K_Q$, $s = s(Q)$, and set

$$S = E(Q, b) + C_0 + Ks, \quad B = Kb.$$

Thus $S \geq C_0$, $E(Q, b) \leq S$, and $S \geq Ks$. Equation (3.2) gives

$$N(Q) \leq m(Q, b) + (R+1)|b| \leq (R+1)(m(Q, b) + |b|).$$

Consequently $S + |B| \geq KN(Q)/(R+1)$; also $S + |B| \geq K|b|$. Choose, for example, $L = 2C_0(R+1)$. Either of the alternatives in the lemma then implies

$$\frac{K}{S + |B|} \leq \frac{1}{C_0}.$$

Using $S \geq Ks$, we obtain the other required smallness condition,

$$\frac{K^2 s}{S(S + |B|)} \leq \frac{K}{S + |B|} \leq \frac{1}{C_0}.$$

All the hypotheses of Proposition 3.4 hold. Its conclusions, with $S/K = m(Q, b) + f_Q$, are precisely (4.1).

Along a path, $s(Q_i) = A^{-i}s(I)$ and $K_{Q_i} = A^{d-i}K_{Q_d}$. Since $K_{Q_d} \geq a_0$,

$$\sum_{i=0}^d \frac{C_0}{K_{Q_i}} \leq \frac{C_0}{a_0} \sum_{\ell=0}^d A^{-\ell}, \quad \sum_{i=0}^d s(Q_i) = s(I) \sum_{i=0}^d A^{-i}.$$

The geometric-series bound gives (4.2). □

We next allow descent to stop according to the values encountered along a path. A stopping rule is understood to use only the squares and tilts already visited. In particular, a violation first detected on arrival at a child stops the procedure at that child.

Lemma 4.2 (Stopped descent). *Start at I with a prescribed tilt b_I . Whenever a step is made, choose the child tilts by Lemma 4.1, and suppose that its applicability condition holds at every square from which the procedure continues. Stop at a time T , no later than the terminal level. Write $m_i = m(Q_i, b_i)$ for the visited values, including the last value m_T . There is a constant C_{st} , independent of the depth and of the stopping rule, such that*

$$\mathbb{E} \left[\sum_{i=0}^T m_i + \sum_{i=0}^{T-1} |b_{i+1} - b_i| \right] \leq C_{\text{st}}(m(I, b_I) + 1). \quad (4.3)$$

The second sum includes the step producing the last square, even if that square causes the procedure to stop.

Proof. There are only finitely many nodes. At each node from which descent continues, choose one of the assignments furnished by Lemma 4.1. These choices and the stopping rule determine all random variables on the finite probability space of terminal paths. Thus only finite conditional-expectation identities are needed. In expressions with a fixed index, set $m_i = 0$ after stopping; those auxiliary values do not enter the stopped sums.

Let d be the depth below I , put $m_0 = m(I, b_I)$, and set

$$F_I = \frac{A}{A-1} \left(\frac{C_0}{a_0} + s(I) \right), \quad M = \mathbb{E} \sum_{i=0}^T m_i.$$

The event $\{i < T\}$ is determined before the next child is selected. Applying (4.1) on this event and summing over $0 \leq i < d$ gives

$$M - m_0 = \sum_{i=0}^{d-1} \mathbb{E}[\mathbf{1}_{\{i < T\}} m_{i+1}] \leq \frac{1}{2} \mathbb{E} \sum_{i=0}^{T-1} (m_i + f_{Q_i}) \leq \frac{1}{2} M + \frac{1}{2} F_I.$$

All the m_i are nonnegative by Definition 3.1. It follows that $M \leq 2m_0 + F_I$. The tilt estimate in (4.1) similarly yields

$$\mathbb{E} \sum_{i=0}^{T-1} |b_{i+1} - b_i| \leq C_1 \mathbb{E} \sum_{i=0}^{T-1} (m_i + f_{Q_i}) \leq 2C_1 m_0 + 2C_1 F_I.$$

Their sum is at most

$$2(1 + C_1)m_0 + (1 + 2C_1)F_I.$$

Since $s(I) \leq 1$, this is bounded by the right-hand side of (4.3) with a constant depending only on the fixed data. This calculation includes $T = 0$, and it includes both m_T and the incoming increment when the last square violates a stopping condition. $\square$

A large tilt whose remaining excess is small forces the original, untilted growth to be large. We give a separate size threshold and ratio threshold for this statement.

Lemma 4.3 (Large slope forces large growth). *For the fixed L in Lemma 4.1, there are constants $B_* \geq 1$ and $\eta_* > 0$ such that every admissible square with $s(Q) \leq 1$ and $K_Q \geq a_0$ satisfies*

$$|b| \geq B_*, \quad m(Q, b) \leq \eta_* |b| \implies N(Q) > L. \quad (4.4)$$

Proof. If no such pair of thresholds existed, then for each positive integer j there would be a square Q_j and a tilt b_j with

$$|b_j| \geq j, \quad \frac{m(Q_j, b_j)}{|b_j|} \leq \frac{1}{j}, \quad N(Q_j) \leq L.$$

Rescale about the center of Q_j by its side $s_j = s(Q_j)$. In the rescaled coordinates, U_j solves $\Delta U_j + K_j^2 p_j U_j = 0$ on $\mathbb{D}_R$, where $K_j = K_{Q_j}$, $\|p_j\|_\infty \leq C$, and $\|\nabla p_j\|_\infty \leq C s_j$. Set

$$S_j = K_j |b_j|, \quad B_j = K_j b_j, \quad D_j = S_j + |B_j| = 2S_j.$$

The hypotheses of Theorem 2.2 follow from

$$S_j \geq a_0 j \longrightarrow \infty, \quad \frac{K_j}{D_j} = \frac{1}{2|b_j|} \longrightarrow 0, \quad \frac{K_j^2 \|\nabla p_j\|_\infty}{S_j D_j} \leq \frac{C s_j}{2|b_j|^2} \longrightarrow 0.$$

Normalize the tilted measures

$$d\mu_j(x) = c_j e^{-2B_j \cdot x} |U_j(x)|^2 dx$$

to have mass one on $\mathbb{D}_R$. By Lemma 2.1 and Theorem 2.2, a subsequence has a subharmonic profile $V \leq 0$, with the identically minus infinity case initially allowed. The definitions give

$$\frac{\log \mu_j(\mathbb{D}_1)}{2S_j} = -\frac{E(Q_j, b_j)}{S_j} = -\frac{m(Q_j, b_j)}{|b_j|} \rightarrow 0.$$

The upper bound in (2.1) on the compact set $\overline{\mathbb{D}_1}$ therefore implies $\sup_{\overline{\mathbb{D}_1}} V = 0$. In particular the profile is nontrivial and attains its maximum at an interior point of $\mathbb{D}_R$. The strong maximum principle yields $V = 0$ on $\mathbb{D}_R$.

Pass to a further subsequence so that $e_j = b_j/|b_j|$ converges to a unit vector e . Undoing the tilt expresses the untilted growth as

$$\frac{E(Q_j, 0)}{S_j} = \frac{1}{2S_j} \log \int_{\mathbb{D}_R} e^{2S_j e_j \cdot x} d\mu_j(x) - \frac{1}{2S_j} \log \int_{\mathbb{D}_1} e^{2S_j e_j \cdot x} d\mu_j(x).$$

For any fixed $1 < r < R$, bound the first integral from below by its restriction to $\mathbb{D}_r$. The weighted form of Lemma 2.1, applied from below on the open disk $\mathbb{D}_r$ and from above on the compact disk $\overline{\mathbb{D}_1}$, gives

$$\liminf_{j \rightarrow \infty} \frac{E(Q_j, 0)}{S_j} \geq \sup_{\mathbb{D}_r} e \cdot x - \sup_{\overline{\mathbb{D}_1}} e \cdot x = r - 1.$$

Letting $r \uparrow R$ gives a lower bound of $R - 1$, without any claim about mass at the boundary of $\mathbb{D}_R$. On the other hand,

$$0 \leq \frac{E(Q_j, 0)}{S_j} = \frac{N(Q_j)}{|b_j|} \leq \frac{L}{j} \rightarrow 0,$$

a contradiction. □

Proposition 4.4 (Stability of large growth). *There are constants $P \geq 1$ and $\sigma > 0$ with the following property. Suppose a square I and a tilt b_I satisfy*

$$|b_I| \geq P, \quad m(I, b_I) \leq \sigma |b_I|. \quad (4.5)$$

Then at least half the paths from I to the terminal level satisfy $N(Q) > L$ at every visited square Q , including I and the terminal square. Equivalently, these paths occupy a measurable set of area at least $|I|/2$.

Proof. Let B_*, η_* be the thresholds from Lemma 4.3. Choose γ so that

$$0 < \gamma < \frac{1}{2}, \quad \frac{\gamma}{1 - \gamma} \leq \eta_*.$$

Let C_{st} be the constant from Lemma 4.2. Choose

$$0 < \sigma \leq \min \left\{ \frac{\gamma}{2}, \frac{\gamma}{8C_{\text{st}}} \right\}, \quad P \geq \max \left\{ 1, \frac{B_*}{1 - \gamma}, \frac{2L}{1 - \gamma}, \frac{8C_{\text{st}}}{\gamma} \right\}.$$

Write $B_0 = |b_I|$. Starting at I , descend using Lemma 4.1 while

$$|b_i - b_I| < \gamma B_0, \quad m_i < \gamma B_0,$$

and stop at the first failure of either inequality or at the terminal level, whichever comes first. These inequalities hold initially by (4.5). Whenever they hold, we have

$$|b_i| > (1 - \gamma)B_0 \geq B_*, \quad |b_i| > L, \quad \frac{m_i}{|b_i|} < \frac{\gamma}{1 - \gamma} \leq \eta_*.$$

Thus every required step is available and Lemma 4.3 gives $N(Q_i) > L$ at each such square.

Let T be this stopping time. Include in the failure event a violation first produced at the terminal level. If a failure occurs, then either $m_T \geq \gamma B_0$, or the triangle inequality gives $\sum_{i=0}^{T-1} |b_{i+1} - b_i| \geq \gamma B_0$. Consequently Lemma 4.2 and Markov's inequality give

$$\begin{aligned} \mathbb{P}(\text{failure}) &\leq \frac{1}{\gamma B_0} \mathbb{E} \left[\sum_{i=0}^T m_i + \sum_{i=0}^{T-1} |b_{i+1} - b_i| \right] \\ &\leq \frac{C_{\text{st}}}{\gamma} \left(\sigma + \frac{1}{P} \right) \leq \frac{1}{4}. \end{aligned}$$

On the complementary event, descent reaches the terminal level and the two strict inequalities hold at every visited square. All of those squares have $N > L$. This event has probability at least $3/4$, which proves the asserted lower bound of $1/2$. The same reasoning includes a subtree of depth zero. $\square$

5 Persistence and packing

We now bound the average of N at the terminal level. The stopped descent estimate controls the contribution of one excursion of high growth, but a path may enter several excursions. We will bound the total area of their entry squares. The argument has two parts: an entry produces a nearby region on which few later entries can occur, and a geometric packing estimate converts these regions into the required area bound.

Fix the root square Q_0 and the constants A, C_0, C_1 from Section 3. We consider finite subdivision trees whose terminal squares have a common side length and for which $K_Q \geq a_0 > 0$ at every level. All constants in this section may depend on these fixed data, including a_0 , but are independent of the eigenfunction and the depth of the tree. Let L be the threshold in Lemma 4.1, and let P, σ be the constants in Proposition 4.4. Proposition 2.5 and the fixed coordinate volume comparisons give a constant C_{init} such that $N(Q_0) \leq C_{\text{init}}$.

5.1 Entries and persistent nearby regions

Definition 5.1 (Excursions). Choose a threshold $H > \max\{L, C_{\text{init}}\}$. Along each descending grid path, an excursion starts the first time $N(Q) > H$ and ends the first subsequent time $N(Q) \leq L$. After an excursion has ended, apply the same rule again. A square at which an excursion starts is called an *entry*. An excursion is *active* at a square if it has started and has not yet ended there.

These decisions depend only on a square and its ancestors, so each entry is a whole grid square. Write $\mathcal{E}$ for the family of entries. The root is not an entry. In addition to the lower bound defining an entry, we have a uniform upper bound on its growth.

Lemma 5.2 (Growth at an entry). *For all sufficiently large fixed H , every entry Q and its grid parent Q^p satisfy*

$$N(Q^p) \leq H, \quad H < N(Q) \leq A^2 \left[\frac{1}{2} + (R+1)C_1 \right] \left(H + \frac{C_0}{a_0} \right). \quad (5.1)$$

Proof. Immediately before an entry the path is inactive. An inactive square cannot have $N > H$, since that would start an excursion, so $N(Q^p) \leq H$. Apply Proposition 3.4 at Q^p with $b = 0$ and $S = HK_{Q^p} + C_0$. Indeed, $E(Q^p, 0) \leq HK_{Q^p} \leq S$, and

$$\frac{K_{Q^p}}{S} \leq \frac{1}{H}, \quad \frac{K_{Q^p}^2 s(Q^p)}{S^2} \leq \frac{1}{H^2},$$

since $s(Q^p) \leq 1$. Thus (3.4) holds when H is sufficiently large. All terms in the averages in (3.5) are nonnegative. There are A^2 children, so the tilt b_Q assigned to the particular child Q satisfies

$$m(Q, b_Q) \leq \frac{A^2 S}{2K_{Q^p}}, \quad |b_Q| \leq \frac{A^2 C_1 S}{K_{Q^p}}.$$

Combining these inequalities with (3.2) and $K_{Q^p} \geq a_0$ proves the upper bound in (5.1). $\square$

For a square Q and a number $D \geq 1$, write DQ for its concentric dilation by the factor D . The following proposition supplies a region of controlled future entry count near each entry. Its location need not be inside the entry square.

Proposition 5.3 (Persistence near an entry). *There are a finite threshold H , a number $\delta > 0$, and an integer $J \geq 0$ such that every entry Q admits a measurable set $Y_Q \subset D_* Q$, where $D_* = 20A$, with*

$$\begin{aligned} |Y_Q| &\geq \delta |Q|, \\ \#\{F \in \mathcal{E} : s(F) < s(Q), y \in F\} &\leq J \quad \text{for almost every } y \in Y_Q. \end{aligned} \quad (5.2)$$

Only entries of the given tree are counted, including when Y_Q extends outside the root.

Proof. The contradiction argument below extracts a nonconstant profile, finds a nearby square with small affine excess relative to a nonzero slope, and transfers that square to Proposition 4.4.

We may require H to exceed any fixed threshold needed above. Suppose there is no triple (H, δ, J) with the asserted uniform property. Choose $H_j \rightarrow \infty$, all above these fixed thresholds and at least 2, and apply this negation with $\delta_j = 1/H_j$ and $J_j = \lceil H_j \rceil$. For each j there are an eigenfunction, a finite tree, and an entry Q_j for which (5.2) fails with these parameters.

There must be more than J_j levels strictly below Q_j . Otherwise the finer-entry count would be at most J_j everywhere, and $Y_{Q_j} = Q_j$ would satisfy (5.2). In particular, every fixed number of further levels is eventually available.

Rescale about the center x_j of the parent Q_j^p , using its side $s_j = s(Q_j^p)$ as unit length. Write k_j for the eigenfrequency of u_j , and put

$$K_j = k_j s_j, \quad S_j = H_j K_j, \quad U_j(\xi) = u_j(x_j + s_j \xi), \quad p_j(\xi) = p(x_j + s_j \xi).$$

The rescaled equation is $\Delta U_j + K_j^2 p_j U_j = 0$. Its coefficients satisfy $\|p_j\|_\infty \leq C$ and $\|\nabla p_j\|_\infty \leq C s_j$. Moreover,

$$S_j \geq H_j a_0 \longrightarrow \infty, \quad \frac{K_j}{S_j} = \frac{1}{H_j} \longrightarrow 0, \quad \frac{K_j^2 \|\nabla p_j\|_\infty}{S_j^2} \leq \frac{C s_j}{H_j^2} \longrightarrow 0.$$

Normalize $|U_j|^2 d\xi$ to have mass one on $\mathbb{D}_R$. Lemma 2.1 and Theorem 2.2, with zero tilt, give a subharmonic profile V on $\mathbb{D}_R$ satisfying

$$V \leq 0, \quad \sup_{\mathbb{D}_1} V \geq -1.$$

For the second assertion, the parent inequality gives $E(Q_j^p, 0)/S_j \leq 1$, and the compact upper bound in (2.1) applies to the closed unit disk. Thus V is not identically $-\infty$. Its supremum on every nonempty open disk in $\mathbb{D}_R$ is therefore finite.

There are only A^2 possible positions of the entry inside its parent. After a subsequence its rescaled center is a fixed point y_e . The entry inequality and the compact upper and open lower bounds in (2.1) imply

$$\frac{1}{A} \leq \liminf_j \frac{E(Q_j, 0)}{S_j} \leq \limsup_j \frac{E(Q_j, 0)}{S_j} \leq \sup_{\mathbb{D}_{R/A}(y_e)} V - \sup_{\mathbb{D}_{1/A}(y_e)} V.$$

The finiteness of the second supremum justifies the subtraction of these bounds. Since $A > 10R$, both disks are contained in $\mathbb{D}_2$. Consequently V is nonconstant on $\mathbb{D}_2$.

The representation in Lemma 3.2 implies $V \in W^{1,1}(\mathbb{D}_2)$: the gradient of the logarithmic kernel is locally integrable in two dimensions, and the representing measure has finite mass. There is a Lebesgue point $z \in \mathbb{D}_2$ of ∇V for which $c = \nabla V(z) \neq 0$. Indeed, otherwise the weak gradient would vanish almost everywhere on the connected disk, forcing V to be constant almost everywhere; its subharmonic representative, which is determined by local averages, would then be constant everywhere.

Extend the fine subdivision grids in the rescaled parent coordinates to the whole plane. For $r = A^{-m}$, choose a grid square with center y_m such that $|y_m - z| \leq r$. These grids have edges $-\frac{1}{2} + nA^{-m}$ in each coordinate and are the same for every j . We claim that

$$\sup_{\mathbb{D}_{Rr}(y_m)} (V(x) - c \cdot x) - \sup_{\mathbb{D}_r(y_m)} (V(x) - c \cdot x) = o(r) \quad (m \rightarrow \infty). \quad (5.3)$$

To prove this, let $W(x) = V(x) - c \cdot x$, and let a_r be its average on $B_r = \mathbb{D}_{(2R+2)r}(z)$. For large m these disks are contained in $\mathbb{D}_2$. Poincaré's inequality and the Lebesgue-point property give

$$\text{avg}_{B_r} |W - a_r| \leq C_R r \text{ avg}_{B_r} |\nabla V - c| = o(r).$$

If $x \in \overline{\mathbb{D}_{Rr}(y_m)}$, the disk $\mathbb{D}_{(R+1)r}(x)$ lies in B_r . The submean inequality for the subharmonic function W therefore gives, uniformly in such x ,

$$W(x) \leq a_r + 4 \text{ avg}_{B_r} |W - a_r| = a_r + o(r).$$

On the other hand,

$$\sup_{\mathbb{D}_r(y_m)} W \geq \text{avg}_{\mathbb{D}_r(y_m)} W \geq a_r - (2R+2)^2 \text{ avg}_{B_r} |W - a_r| = a_r - o(r).$$

This proves (5.3). Subharmonicity is essential here: it turns the average control into a uniform upper bound.

Now fix $m \geq 2$ sufficiently large that $Rr \leq 1/4$ and the quotient of the left side of (5.3) by $r|c|$ is strictly less than σ . In the original coordinates let I_j be the square with center $x_j + s_j y_m$ and

side $s_j r$, and assign it the tilt $b_{I_j} = H_j c$. The additional weight in parent coordinates is $e^{-2S_j c \xi}$, up to a constant canceled in the norm ratio. The weighted bounds of Lemma 2.1 therefore yield

$$\limsup_j \frac{m(I_j, b_{I_j})}{|H_j c|} \leq \frac{\sup_{\mathbb{D}_{Rr}(y_m)}(V(x) - c \cdot x) - \sup_{\mathbb{D}_r(y_m)}(V(x) - c \cdot x)}{r|c|} < \sigma. \quad (5.4)$$

The factor is exactly $1/(r|c|)$, since $K_{I_j} = rK_j$ and $S_j = H_j K_j$. Because $c \neq 0$, we also have $|H_j c| \geq P$ for all large j . Thus (4.5) holds at I_j .

We verify that Proposition 4.4 can be used on the whole subtree of I_j down to the original bottom. The chosen m is fixed before j tends to infinity, so that level is eventually available. All squares of a given extended-grid level have the same physical scale as the original tree, and hence have $K \geq a_0$. The center y_m stays in $\mathbb{D}_3$, and $Rr \leq 1/4$. The disks required for I_j and all its descendants are consequently contained in $\mathbb{D}_{5s_j}(x_j)$: even the root margin has rescaled radius $2Rr$, and $|y_m| + 2Rr < 3 + 1/2 < 5$. Since $Q_j^p \subset Q_0$, we have $|x_j - x_{Q_0}| \leq s_0/\sqrt{2}$ and $s_j \leq s_0$, so this disk lies in $\mathbb{D}_{6s_0}(x_{Q_0}) \subset \mathbb{D}_{2Rs_0}(x_{Q_0})$, within the fixed coordinate patch. This also covers the larger disk margins required when regarding I_j as the root of a subtree.

Proposition 4.4 now gives a measurable subset of I_j of area at least $|I_j|/2$ on which every visited square from I_j through the bottom has $N > L$. The extended grids align with the original root: a root edge differs from the corresponding parent edge by an integer multiple of s_j , and the extended parent grid has rescaled edges $-1/2 + nA^{-m}$. Thus a fine grid square is either contained in Q_0 or has interior disjoint from it. In the latter case its points belong to no entry of the original tree. In the former case its descendants are exactly those of the original tree. From the level of I_j onward there can be at most one entry along any of these paths, since two starts require an intervening value $N \leq L$. Before that level there are at most m relevant levels below the tested entry. Thus the number of strictly finer entries is at most $m + 1$ on this subset.

Finally, the subset lies in $20AQ_j$: in parent units that dilated square has half-side 10, whereas $y_m \in \mathbb{D}_3$ and the entry center lies in the unit parent square. Its area relative to the entry is at least

$$\frac{|I_j|}{2|Q_j|} = \frac{1}{2}A^{2-2m} > 0.$$

Both this positive number and $m + 1$ are fixed independently of j . They eventually satisfy the tested requirements $\delta_j = 1/H_j$ and $J_j = \lceil H_j \rceil$, contradicting the choice of Q_j . This proves the proposition. $\square$

5.2 Packing with witnesses outside the squares

The next geometric lemma converts the nearby regions just constructed into a bound on the sum of entry areas. We state it for an arbitrary finite family in a subdivision tree.

Lemma 5.4 (Packing from nearby witnesses). *Let $\mathcal{F}$ be a finite family of squares in the A -adic subdivision tree of a square Q_0 , where $A \geq 2$. Suppose there are fixed $D \geq 1$, $\delta > 0$, and an integer $J \geq 0$ such that every $Q \in \mathcal{F}$ admits a measurable set $Y_Q \subset DQ$ satisfying*

$$|Y_Q| \geq \delta|Q|, \quad \#\{F \in \mathcal{F} : s(F) < s(Q), y \in F\} \leq J \quad \text{for almost every } y \in Y_Q.$$

Then

$$\sum_{Q \in \mathcal{F}} |Q| \leq C(A, D, \delta, J)|Q_0|. \quad (5.5)$$

Proof. Ignore grid boundaries and the boundaries of the finitely many dilates DQ with $Q \in \mathcal{F}$; these form a null set. Define

$$n(Q) = 1 + \#\{F \in \mathcal{F} : Q \subsetneq F\}.$$

Squares of a fixed rank n have disjoint interiors. At each rank, process the squares in decreasing order of side length, retaining a square if its D -dilate has interior disjoint from the previously retained dilates. Let $\mathcal{S}$ be the union of these selected families. Every discarded square Q has its dilate meeting that of a selected square P with $s(P) \geq s(Q)$; hence $Q \subset (2D+1)P$. Covering the original disjoint squares at that rank by these larger dilates and then summing over ranks gives

$$\sum_{Q \in \mathcal{S}} |Y_Q| \geq \delta \sum_{Q \in \mathcal{S}} |Q| \geq \alpha \sum_{Q \in \mathcal{F}} |Q|, \quad \alpha = \frac{\delta}{(2D+1)^2}.$$

The selected witness sets at a single rank are disjoint up to null sets. It remains to control overlap between different ranks. For a point y outside the ignored boundaries, put

$$n_{\text{tot}}(y) = \#\{F \in \mathcal{F} : y \in F\}.$$

If a witness point y lies inside its square Q , every larger square containing y is an ancestor of Q , so the witness assumption makes $n_{\text{tot}}(y)$ differ from $n(Q)$ by at most J . An exterior witness point can also disagree with the ancestry of Q at larger scales. For $Q \in \mathcal{S}$ and $y \in Y_Q$, let

$$d_Q(y) = \sum_{\substack{F \in \mathcal{F} \\ s(F) > s(Q)}} \left| \mathbf{1}_F(y) - \mathbf{1}_{\{Q \subset F\}} \right|$$

count the mismatches between membership of y in a larger square and that square's being an ancestor of Q . If b is the number of larger squares in $\mathcal{F}$ containing y , then

$$|n(Q) - 1 - b| \leq d_Q(y).$$

At most one square of side $s(Q)$ contains y , and the witness assumption bounds the number of smaller squares containing it by J . Decomposing $n_{\text{tot}}(y)$ into these three size ranges gives

$$|n(Q) - n_{\text{tot}}(y)| \leq d_Q(y) + J + 1 \quad \text{for almost every } y \in Y_Q. \quad (5.6)$$

Thus a uniform bound for $d_Q(y)$ restricts the ranks that can contribute at y to a fixed interval. We next show that the total witness area weighted by d_Q is controlled:

$$\sum_{Q \in \mathcal{S}} \int_{Y_Q} d_Q(y) dy \leq B \sum_{F \in \mathcal{F}} |F|, \quad (5.7)$$

where B depends only on A, D .

Fix $F \in \mathcal{F}$ and a finer side length $t = A^{-\ell} s(F)$, $\ell \geq 1$. The grid is laminar: each square Q of side t is either contained in F or interior-disjoint from it. If it can contribute a mismatch, a point of DQ lies on the opposite side of ∂F from Q . It follows that the whole square Q lies in the collar

$$\{x : \text{dist}_\infty(x, \partial F) \leq (D+1)t\},$$

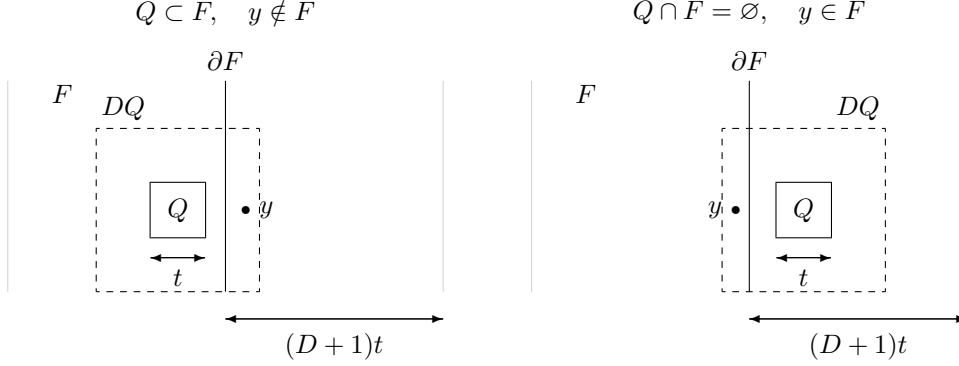


Figure 1: Ancestry and spatial membership can disagree in either direction. Each panel shows one witness point $y \in Y_Q \subset DQ$ and a fine square of side $t = s(Q)$ near a face of the larger square F . The dashed square is DQ ; the pale lines mark the boundary of the collar of width $(D+1)t$ on each side of the face.

where dist_∞ is distance for the maximum norm on $\mathbb{R}^2$. Figure 1 shows the two possible directions of this mismatch near one face of F . This collar has area at most

$$8(D+1)s(F)t + 4(D+1)^2t^2.$$

All squares of side t , across all ranks, have disjoint interiors. Since $|Y_Q| \leq D^2|Q|$, their total contribution for this fixed F and t is at most D^2 times that collar area. Summing the geometric series over $\ell \geq 1$ and then summing over F proves (5.7). For example, one may take

$$B = D^2 \left(\frac{8(D+1)}{A-1} + \frac{4(D+1)^2}{A^2-1} \right).$$

Choose an integer $M \geq 1$ with $M \geq 2B/\alpha$, and set $Z_Q = \{y \in Y_Q : d_Q(y) \leq M\}$. Remove also the null sets where the witness assumption fails. Markov's inequality and (5.7) give

$$\sum_{Q \in \mathcal{S}} |Z_Q| \geq \frac{\alpha}{2} \sum_{Q \in \mathcal{F}} |Q|.$$

For $y \in Z_Q$, Equation (5.6) now gives

$$|n(Q) - n_{\text{tot}}(y)| \leq M + J + 1.$$

Only $2(M + J + 1) + 1$ ranks can therefore contribute at y . At a given rank the selected witness sets are disjoint, so this is also a bound on the total overlap of the Z_Q .

For $D \geq 1$, the inclusions $Q \subset Q_0$ imply $DQ \subset DQ_0$. Integrating the overlap bound over this common region yields

$$\frac{\alpha}{2} \sum_{Q \in \mathcal{F}} |Q| \leq (2(M + J + 1) + 1)D^2|Q_0|.$$

This proves (5.5). $\square$

Apply Lemma 5.4 to the family of entries, using Proposition 5.3 with $D = D_*$. We obtain a constant C such that the sum of all entry areas is at most $C|Q_0|$.

5.3 The terminal mean growth

Theorem 5.5 (Bounded mean growth). *For the finite trees considered in this section, let $\mathcal{T}_{\text{term}}$ be the collection of terminal squares and let s_{term} be their common side length. There is a constant C , independent of the eigenfunction and the number of levels, such that*

$$\text{avg}_{q \in \mathcal{T}_{\text{term}}} N(q) \leq C, \quad \text{avg}_{q \in \mathcal{T}_{\text{term}}} E(q, 0) \leq C k s_{\text{term}}. \quad (5.8)$$

Proof. Fix the threshold H supplied by Proposition 5.3. A terminal path on which no excursion is active has terminal growth $N \leq H$. Every other terminal path has exactly one final active entry. For an entry Q , let $\Omega_Q \subset Q$ be the union of terminal squares on which that entry's excursion is still active. These sets are disjoint as Q varies.

Starting separately at each entry Q , set $b_Q = 0$ and perform the descent of Lemma 4.1, stopping at the first square with $N \leq L$ or at the bottom, whichever comes first. Every continuing square has $N > L$, so the descent is available regardless of its current tilt. The low-growth stopping square, if any, is included as the last produced value; no further step is needed there. These separate choices of tilts do not affect excursion membership, which is defined entirely by the untilted quantity N .

Along a random descendant path from Q , denote the stopping time by T and the successive values by m_i, b_i . Since $b_0 = 0$, (3.2) gives

$$N_T \leq m_T + (R+1)|b_T| \leq m_T + (R+1) \sum_{i < T} |b_{i+1} - b_i|.$$

The uniform distribution on paths agrees with normalized planar area on Q . Lemma 4.2 and Lemma 5.2 therefore imply

$$\begin{aligned} \int_{\Omega_Q} N_{\text{term}}(x) dx &\leq |Q| \mathbb{E} \left[m_T + (R+1) \sum_{i < T} |b_{i+1} - b_i| \right] \\ &\leq C|Q|(N(Q) + 1) \leq C'|Q|. \end{aligned}$$

Here N_{term} is constant on each terminal square, with value N of that square. The first inequality merely restricts a nonnegative integral to the paths still active at the bottom; it does not condition on that event or divide by its probability.

Sum over all entries and use (5.5), then include the inactive terminal paths. This gives

$$\int_{Q_0} N_{\text{term}}(x) dx \leq H|Q_0| + C' \sum_{Q \text{ entry}} |Q| \leq C|Q_0|.$$

Equal terminal areas turn this integral bound into the first inequality of (5.8). The second follows because every terminal square has $K_q = k s_{\text{term}}$ and $E(q, 0) = K_q N(q)$. $\square$

6 Nodal length at the wavelength scale

We first state the planar estimate that relates nodal length to growth. The disks in this section are Euclidean.

Theorem 6.1 (Roy-Fortin). *There are constants $\varepsilon_0 > 0$ and $C > 0$ with the following property. Let F be a nonzero real solution of*

$$\Delta F + qF = 0 \quad \text{on } \mathbb{D}_3, \quad q \in C^\infty(\mathbb{D}_3), \quad \|q\|_\infty < \varepsilon_0.$$

If

$$\beta = \log \frac{\|F\|_{L^\infty(\mathbb{D}_{5/2})}}{\|F\|_{L^\infty(\mathbb{D}_{1/4})}},$$

then

$$\mathcal{H}^1(Z_F \cap \mathbb{D}_{1/60}) \leq C \max\{1, \beta\}.$$

This is [11, Theorem 2.1.1]. The potential need not have a constant sign. We use the following consequence, keeping the norm and length scales explicit.

Lemma 6.2 (Nodal length from local L^2 growth). *Fix the coefficient bounds and coordinate neighborhood from Section 3. There is an $\varepsilon > 0$ such that, for every admissible square I of side s with $ks \leq \varepsilon$,*

$$\mathcal{H}^1(I \cap Z_u) \leq Cs(1 + E(I, 0)). \quad (6.1)$$

The constants are independent of u , k , and s .

Proof. Let x_I be the center of I , and put

$$F(z) = u(x_I + 60sz), \quad q(z) = (60ks)^2 p(x_I + 60sz).$$

The disk of physical radius $180s$ is available because $R = 1000$. Choose ε so small that $(60\varepsilon)^2 \|p\|_\infty < \varepsilon_0$ uniformly over the coordinate regions in use. Then F satisfies Theorem 6.1.

An interior H^2 estimate for $\Delta F = -qF$, followed by the two-dimensional Sobolev embedding, gives

$$\|F\|_{L^\infty(\mathbb{D}_{5/2})} \leq C \|F\|_{L^2(\mathbb{D}_{11/4})}.$$

Only the uniform bound on q is needed here. On the smaller disk,

$$\|F\|_{L^2(\mathbb{D}_{1/4})} \leq |\mathbb{D}_{1/4}|^{1/2} \|F\|_{L^\infty(\mathbb{D}_{1/4})}.$$

Consequently

$$\beta \leq C + \log \frac{\|F\|_{L^2(\mathbb{D}_{11/4})}}{\|F\|_{L^2(\mathbb{D}_{1/4})}}.$$

The same L^2 version of the local length bound is recorded directly in [12, Theorem 3.1.1].

In physical coordinates the two norm radii are $165s$ and $15s$. Their common change-of-variables factor cancels, and nesting gives

$$\log \frac{\|F\|_{L^2(\mathbb{D}_{11/4})}}{\|F\|_{L^2(\mathbb{D}_{1/4})}} = \log \frac{\|u\|_{L^2(\mathbb{D}_{165s}(x_I))}}{\|u\|_{L^2(\mathbb{D}_{15s}(x_I))}} \leq E(I, 0).$$

All these norms are nonzero by Proposition 2.5. The length disk becomes $\mathbb{D}_s(x_I)$ and contains the closed square I , whose corner distance is $s/\sqrt{2}$. Hausdorff length scales by $60s$. Theorem 6.1 therefore yields (6.1). $\square$

Proof of Theorem 1.1. Choose finitely many smooth isothermal neighborhoods and root squares Q_0 whose interiors cover M . Make their sides $s_0 \leq 1$ small enough that each closed disk of radius $2Rs_0$ about a root center is contained in its coordinate neighborhood. The conformal factors and their first derivatives have uniform bounds on the finitely many regions used below, and the conformal factors are bounded away from zero there. Proposition 2.5 gives a uniform initial bound $N(Q_0) \leq C_{\text{init}}$.

Choose the subdivision constants from Proposition 3.4, and then the small ε of Lemma 6.2. Write $k_{\min} = \sqrt{\lambda_1(M, g)} > 0$. A single permissible lower scale bound for all roots is

$$a_0 = \min \left\{ \frac{\varepsilon}{A}, k_{\min} \min_{Q_0} s(Q_0) \right\} > 0.$$

Now fix the descent and persistence constants with this a_0 . This order of choices makes every constant depend only on the fixed metric and cover.

For a given eigenfunction, subdivide each root to the first level at which $ks \leq \varepsilon$, where s is that level's common side. If this level is below the root, its parent has $kAs > \varepsilon$, so $ks > \varepsilon/A$. If it is the root, then $ks \geq k_{\min}s_0$. In both cases, every visited square satisfies $K_Q \geq a_0$. Theorem 5.5 therefore gives

$$\text{avg}_I \text{ terminal in } Q_0 E(I, 0) \leq Cks.$$

There are $(s_0/s)^2$ terminal squares. Applying Lemma 6.2 and summing their lengths yields

$$\begin{aligned} \mathcal{H}^1(Q_0 \cap Z_u) &\leq Cs \left(\frac{s_0}{s} \right)^2 (1 + Cks) \\ &\leq Cs_0^2(1/s + k) \leq Cs_0^2(a_0^{-1} + 1)k. \end{aligned}$$

Subadditivity suffices even if a nodal arc lies on a grid edge.

On each convex coordinate square, the inverse coordinate map is Lipschitz into (M, d_g) with constant at most $\sqrt{\sup p}$: the coordinate line segment is an admissible path for the Riemannian distance. Hence the preceding Euclidean Hausdorff-length bound gives the same bound for $\mathcal{H}_g^1$ up to a fixed factor. Sum over the finite root cover to obtain $\mathcal{H}_g^1(Z_u) \leq C(M, g)k$. The coordinate choices and all constants were independent of the eigenfunction and of its eigenspace. Since $k = \sqrt{\lambda}$, this proves the theorem. $\square$

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